MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (Honours)/ MGU-BBA (Honours)/ MGU-BCA (Honours)

REGULAR EXAMINATION October 2026

Fifth SEMESTER

Discipline Specific Elective (DSE) - MG5DSEECT310 - ROBOTICS AND INDUSTRIAL AUTOMATION

(2024 ADMISSION ONWARDS)

QP Code: 26900408

Valuation Scheme

Duration: 1 hour 30 minutes Maximum Marks: 50

Part A – Short Answer Questions (2 marks each)

Answer any 10 questions. Award 2 marks for each correct response. (10 × 2 = 20)

1. What are the main objectives of using robots in industrial applications?
   Ans: Productivity, accuracy, safety, consistency; any two.
2. What is a geared DC motor?
   Ans: DC motor coupled to a gearbox to reduce speed and increase torque.
3. What is meant by position control in a servo motor?
   Ans: Controlling and maintaining the angular/linear position using feedback.
4. List any two important features of Industry 4.0.
   Ans: IoT connectivity, automation, cyber-physical systems, real-time data; any two.
5. What is a transistor output module?
   Ans: PLC output module using semiconductor switching for DC loads.
6. What is a PLC input?
   Ans: Signal received by PLC from a switch, sensor or other field device.
7. What is an alarm output in a sensor-based PLC system?
   Ans: Warning signal activated when a sensed condition exceeds a set limit.
8. What is the role of an output status indicator?
   Ans: Shows whether a PLC output is ON or OFF.
9. What is a coil in Ladder Logic?
   Ans: Output instruction representing an energized output or internal relay bit.
10. What is interlocking in motor control?
    Ans: Logic preventing conflicting or unsafe motor operations.
11. List any two applications of control systems in robotics.
    Ans: Position control, speed control, trajectory tracking, force control; any two.
12. State one advantage of robotic spot welding.
    Ans: High repeatability, consistent weld quality, speed or safety; any one.
13. What is a pick-and-place robot?
    Ans: Robot that picks an object from one location and places it at another.

14. What is the role of a controller in an automatic water level system?
Ans: Monitors level sensor and switches pump/valve to maintain desired level.

Part B – Short Essay Questions (5 marks each)

Answer any 6 questions. Award marks for relevant explanation, diagram and examples as indicated. (6 × 5 = 30)

15. Explain the important features of a stepper motor that make it suitable for position control applications.
Ans: Stepwise rotation and pulse control (2); accuracy/repeatability (1); holding torque (1); applications (1).

16. Explain the basic working principle of a piezoelectric sensor and mention its applications.
Ans: Piezoelectric effect (2); construction/operation (1); applications such as vibration/pressure/impact (2).

17. Explain the role of electrical actuators in robotic and automated systems.
Ans: Definition and electrical-to-mechanical conversion (1); types/examples (2); roles in movement/control (2).

18. Design and explain a start-stop latching circuit for a motor using Ladder Logic.
Ans: Correct ladder diagram with stop NC, start NO, seal-in contact and coil (3); operation (2).

19. Explain how a PLC can monitor gas concentration and generate an alarm when a preset limit is exceeded.
Ans: Gas sensor and PLC input/interface (2); comparison with setpoint (1); alarm output/logic (2).

20. Explain the basic working principles of temperature sensor and its PLC interfacing requirements.
Ans: Temperature sensing principle/examples (2); signal conditioning and PLC input (2); scaling/monitoring (1).

21. Explain the major components of an automatic conveyor belt mechanism and state the function of each.
Ans: Belt/rollers/frame (2); motor and drive (1); sensors/controller (1); functions and integration (1).

22. Explain the importance of an automatic conveyor belt mechanism in industrial material handling.
Ans: Continuous transport and throughput (2); reduced labour and time (1); safety/consistency (1); industrial applications (1).